NETWORKING AND SERVICES

Basic Administration

BASIC ADMINISTRATION CAPABILITIES

- Monitoring and managing disk space
- Managing user groups and accounts
- Managing file and folder permissions
- Managing processes
- Scheduling tasks
- Configuring network settings and services

NETWORK CONFIGURATION

NETWORK CONFIGURATION FILES

In Ubuntu, network configuration is determined by files:

- **`/etc/hostname`** – for setting the hostname of the local machine
- **`/etc/netplan/*.yaml`** - In modern Ubuntu, network configuration (including IP addresses, netmasks, and gateways) is determined by YAML files. This replaces the older `/etc/network/interfaces` and `/etc/resolv.conf`.

NETWORK CONFIGURATION FILES

- Sample network configuration yaml file:

```
network:
  version: 2
  ethernets:
    eth0:
      dhcp4: no
      addresses: [192.168.0.2/24]
      routes:
        - to: default
          via: 192.168.0.1
      nameservers:
        addresses: [192.168.0.254, 8.8.8.8]
```

NETWORK CONFIGURATION FILES

In Ubuntu, network configuration is determined by files:

- **/etc/hosts** – lists hostnames and corresponding address to supplement DNS

  E.g. `127.0.0.1  localhost`

- **/etc/hosts.allow** and **/etc/hosts.deny**– lists services, hosts and corresponding permission to access

  E.g. `sshd: 192.168.1.25 : allow`

NETWORK CONFIGURATION COMMANDS

- `ip addr`
  - Replaces `ifconfig`. Can be used to view interface network settings and IP configuration
- `ip route`
  - Replaces `route`. Used to manage network routes, commonly used to view or set the default gateway.
- `netplan try`
  - Parses the configuration and applies it temporarily. If the administrator does not confirm within a timeout (saving them from being locked out over SSH), it automatically rolls back.
- `netplan apply`
  - Replaces older `initctl` scripts. Used to apply the new Netplan configuration, restart the network service, and reset interfaces to the saved YAML configurations.

NETWORK SERVICES

NETWORK SERVICES

- Server systems are expected to host a form of network service needed by an organization
  - File and print services
  - Web services
  - Email services
  - Remote access services
  - Host resolution and IP address services
- Daemons
  - Services that run in the background which end users do not directly control
  - "Listen" for requests and exchange messages and data as requested by clients

CENTRALIZATION AND DECENTRALIZATION

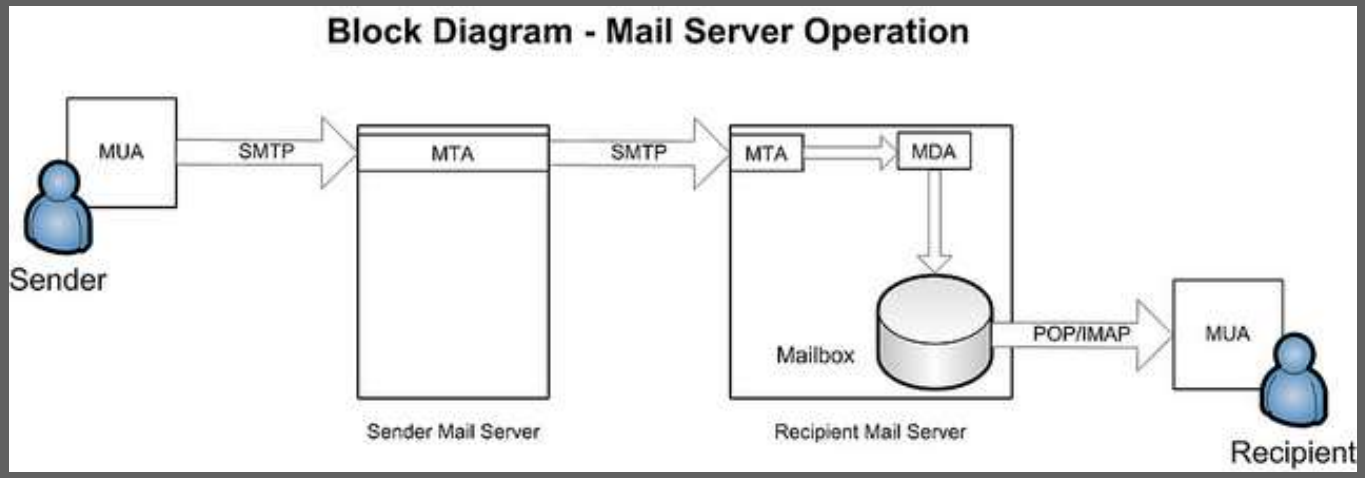
- **Centralization**
  - Refers to pulling groups together and consolidating duplicate systems
  - Reasons to centralize
    - Service consolidation into fewer hosts
    - System administration and administrative staff expertise
    - Infrastructure decisions
    - Technology that became a commodity
- **Decentralization**
  - Distributing systems for more autonomy
  - Reasons to decentralize
    - Fault tolerance
    - Customization specific to a group

FILE AND PRINT SERVICES

- **Server Message Block (SMB)**
  - Service that provides shared access to printers, folder and files among hosts within a local network
  - Establishes a long term connection where a shared resource can be accessed similarly to a local resource on a client
  - Implementations
    - Common Internet File Service (CIFS) – Microsoft
    - SAMBA – UNIX and Linux systems
- **Network File Service (NFS)**
  - A distributed file system service that allows file access over the network
  - Commonly used by UNIX / Linux clients to mount a remote filesystem or portion of it
- **Considerations:**
  - Access policies for both files and printers

EMAIL SERVICES

- **Mail User Agent (MUA):** Component which interacts with end users directly
- **Mail Transfer Agent (MTA):** Transfers an email from a sending mail server to a recipient mail server. (e.g. Postfix, Sendmail)
- **Mail Delivery Agent (MDA):** Delivers email received by the local MTA to a user's mailbox



http://xmodulo.com/

EMAIL SERVICES

- **Sending email:** Simple Mail Transfer Protocol (SMTP)
- **Receiving email:**
  - Post Office Protocol (POP) - download and delete method
  - Internet Message Access Protocol (IMAP) – allows mail to be managed from multiple devices
- **Considerations**
  - **Privacy and usage policies** – generally not encrypted
  - **Namespaces** – Have a default token then let users choose. Use a directory service to support lookup
  - **Filtering** – Best practice to do this at the server
  - **Scaling –** needs to scale with users and with time
  - **Simplicity** – Limit number of machines involved

WEB SERVICE

- Basic components are the browsers, web service and URL
- Key roles:
  - Webmaster – manages web site content
  - Web System Administrator – maintains the hardware and software that make up the web server
- Considerations
  - Content management– Who and when?
  - Scaling – horizontal (replicate then load balance) or vertical (separate service to multiple components)
  - Security – content and application protection
  - Technology – A popular one is LAMP (Linux, Apache, MySQL, PHP/Python/Perl)

REMOTE ACCESS SERVICE

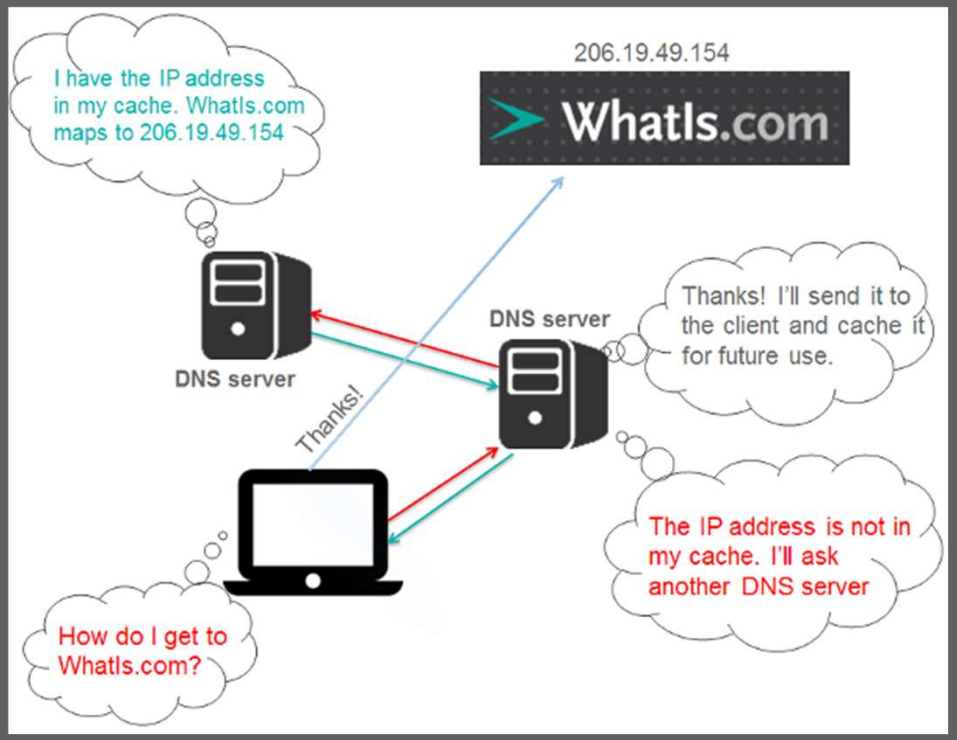
- Caters to people who need to access their work away from the system
- Secure Shell (SSH)
  - Provides an encrypted method of access to a text-based terminal interface (VTY)
  - Uses public key cryptography to authenticate systems and users
- Considerations:
  - Permitted accounts – Don't allow 'root'
  - Permitted remote hosts
  - Authentication method – public key authentication is preferred over password authentication

REMOTE ACCESS: SSH WITH CERTIFICATES

- **Secure Shell (SSH):** Provides an encrypted method to access a text-based terminal interface remotely.
- **Certificate-Based Authentication**
  - Instead of relying solely on standard public key cryptography, systems can be configured to use SSH certificates.
  - A trusted Certificate Authority (CA) signs a user's or host's public key, creating a certificate.
- **Management Advantage**
  - Simplifies administration for permitted remote hosts; servers only need to trust the CA rather than storing individual public keys for every user

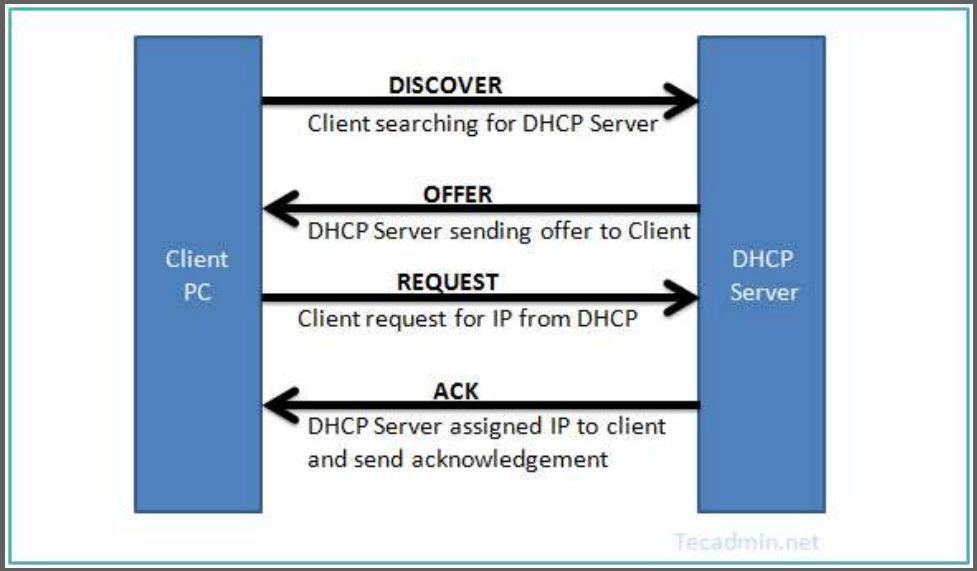
DOMAIN NAME SERVICE

- Service that performs resolution of hostnames to IP addresses
- Entries are stored as different record types
  - A – end device
  - MX – domain mail server
  - NS – authoritative name server
  - CNAME – canonical name for hosts that have several aliases
- Considerations
  - Security – DNS poisoning and DoS
  - Caching timeout
  - Placement in the network
  - Capacity – 100 bytes RAM per entry



IP ADDRESSING SERVICE

- Dynamic Host Configuration Protocol (DHCP) provides a service that allows automatic network configuration of hosts
- Considerations:
  - Lease Time - based on user turnover and infrastructure capacity
  - Pool size – based on infrastructure capacity



http://tecadmin.net

SOFTWARE DEPOT SERVICE

- A software depot is a method of making software packages available to users
  - On Windows systems – usually in the form of repositories from where software can be installed
  - On UNIX systems – usually in the form of repositories of software that can be mounted and used in real time over the network

- Considerations
  - Policy for contributing to repository
  - Updating
  - Licensing for commercial software

BEST PRACTICES: SYSTEM HARDENING

- System hardening is a crucial best practice in system administration to secure server systems and network services against vulnerabilities.
  - **Network Access Control**
    - Utilize configuration files like `/etc/hosts.allow` and `/etc/hosts.deny` to explicitly define which hosts have permission to access specific services.
  - **Remote Access Hardening**
    - For SSH, do not permit the 'root' account to log in directly, and mandate public key or certificate authentication over passwords.
  - **File Service Security**
    - Implement strict access policies for both shared files and printers on services like CIFS or NFS.
  - **DNS Protection**
    - Implement safeguards against specific threats like DNS poisoning and Denial of Service (DoS) attacks.

MAINTAINING SERVICES

SYSTEMATIC DIAGNOSIS AND TROUBLESHOOTING

- **Learn the Problem**
  - Understand the customer's specific issue with the network service (e.g., Web, DNS, DHCP) before acting.
- **Process of Elimination**
  - Systematically remove components or isolate segments of the network until the problem disappears.
- **Successive Refinement**
  - Gradually add components back into the system and verify the desired change or functionality.
- **Use the Right Tools:**
  - Utilize command-line utilities for systematic diagnosis, such as `ifconfig` or `ip addr` to view interface settings or route to inspect network routes.
- **Fix the Cause**
  - Address the root issue of the service failure rather than just the symptoms and avoid temporary fixes.

CHANGE PROCESSES: CONVERSIONS AND UPGRADING

- **Service Changing:** Involves the strategic transition from an existing IT service to a new or alternative solution, requiring careful scheduling to minimize organizational disruption.
- **Conversions:** The process of adapting or migrating data, user accounts, and system configurations to be compatible with a new system architecture.
- **Upgrading:** Implementing newer versions of software or hardware components to improve system performance, security, or to add features.
- **Architectural Impact:** These changes often align with infrastructure decisions regarding centralization (consolidating services) or decentralization (distributing systems for fault tolerance).

CHANGE MANAGEMENT

- Process that ensures effective planning, implementation and post-event analysis of any changes to a system
- Components:
  - Communication and scheduling – proper scheduling and inform those who are affected
    - Routine Update – small scope, low risk (no need to communicate)
    - Major Update – Large scope or requires outage (push communication and schedule during off-peak or maintenance windows)
    - Sensitive Update - May cause outage (post communications and schedule during off-peak))
  - Planning and testing – plan how to proceed and test before proceeding
  - Process and documentation – well documented and approved process
  - Revision control and automation – track changes and back out if issues arise

MAINTENANCE WINDOWS

- Period in which a lot of systems work must be performed and is disruptive to the rest of the organization
- Why is it needed?
  - Clean out systems and perform any hardware changes
  - Consolidate downtime to a schedule that causes the least disruption

Preparation

- Scheduling
- Change proposals
- Master planning

Execution

- Blocking access
- Execution
- Testing

Resolution

- Announcing completion
- Enable Access
- Standby for issues

MAINTENANCE WINDOWS

- Considerations
  - Scheduling – must be at a time when it is least disruptive
  - Planning – must be far in advance if it involves hardware
  - Directing – coordination of communications and tasks
  - Master plan – ensure that change proposals do not conflict with each other and will fit in the window
  - Deadline for change completion – tracking change progress and deciding when to abort a change

NETWORKING AND SERVICES

Basic Administration